

Notes on the Resolution of the Spielman–Teng Conjecture

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Abstract

The Spielman–Teng conjecture predicted that for an $n \times n$ random matrix M with iid subgaussian entries, $\mathbb{P}(\sigma_n(M) \leq \varepsilon n^{-1/2}) \leq C\varepsilon + e^{-cn}$. The best prior results left a gap: the constant C couldn't be pushed to 1. In 2024, Sah, Sahasrabudhe, and Sawhney proved the conjecture up to a $1 + o(1)$ factor, showing $\mathbb{P}(\sigma_n(M) \leq \varepsilon n^{-1/2}) \leq (1 + o(1))\varepsilon + e^{-\Omega(n)}$. These notes walk through their proof architecture and verify the bound computationally.

1 Introduction

I first encountered the Spielman–Teng conjecture in Zixiang's office hours, scribbled on the corner of a whiteboard that hadn't been fully erased. He was talking about smoothed analysis and why the simplex method works so well in practice. The question on the board was about the smallest singular value of a random matrix. How small can it be?

The answer, it turns out, has a long history. For an $n \times n$ matrix A , the condition number

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

governs everything from the stability of linear solvers to convergence rates of iterative methods. Worst-case condition numbers can be absurd. The 50×50 Hilbert matrix has $\kappa(H_{50}) > 10^{17}$. But in practice, matrices that arise from real data or random perturbations of structured inputs behave far better.

Spielman and Teng [8] formalized this intuition with smoothed analysis, which asks: given an adversarially chosen matrix M and a small random perturbation σG , how large is $\mathbb{E}[\kappa(M + \sigma G)]$? They conjectured the answer is $O(n/\sigma)$, regardless of the base matrix M . The conjecture sat open for nearly two decades. Partial progress came from Sankar, Spielman, and Teng [7], who proved $O(n^{3/2}/\sigma)$, and from Tao and Vu [9], who got $O(n/\sigma)$ with logarithmic losses.

But the question that mattered more, at least from a random matrix theory perspective, was sharper. Forget smoothed analysis for a moment. Take M to be a random matrix with iid entries. What's the probability that its least singular value $\sigma_n(M)$ is tiny?

Edelman [2] answered this exactly for Gaussian matrices in 1988. The rescaled quantity $n \cdot \sigma_n(M)$ converges in distribution to a random variable with density

$$f(x) = (x + 1)e^{-x^2/2 - x}, \quad x \geq 0. \tag{1}$$

This gives $\mathbb{P}(\sigma_n(G) \leq \varepsilon n^{-1/2}) \approx \varepsilon$ for small ε , where G is Gaussian.

For non-Gaussian entries, Rudelson and Vershynin [4, 5] proved the bound

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) \leq C\varepsilon + e^{-cn}, \quad (2)$$

where $C > 1$ is an absolute constant. Tao and Vu [10] improved this to $C = 1$ for $\varepsilon \geq n^{-c}$, but the full range remained out of reach. Tikhomirov [11] made a breakthrough of a different kind, proving that the singularity probability of a random Rademacher (± 1) matrix is exactly $2^{-n+o(n)}$.

The Spielman–Teng conjecture, in its sharpest form, asked for $C = 1$. Sah, Sahasrabudhe, and Sawhney [6] proved it, up to a $1 + o(1)$ factor. That’s what these notes are about.

The paper is organized as follows. Section 2 collects definitions and tools. Section 3 states the main theorems. Section 4 walks through the four-step proof, which is the heart of these notes. Section 5 describes Monte Carlo experiments that verify the bound empirically. Section 6 discusses what’s left.

2 Preliminaries

2.1 Subgaussian Random Variables

Definition 2.1. A centered random variable ξ is *subgaussian* with parameter K if $\mathbb{E}[e^{t\xi}] \leq e^{K^2 t^2/2}$ for all $t \in \mathbb{R}$. The subgaussian norm is $\|\xi\|_{\psi_2} = \inf\{K > 0 : \mathbb{E}[e^{\xi^2/K^2}] \leq 2\}$.

Standard Gaussians are subgaussian. So are Rademacher random variables (± 1 with equal probability), bounded random variables, and more generally any random variable whose tails decay at least as fast as a Gaussian’s. A detailed treatment is in Vershynin [12], Chapters 2–3.

2.2 Singular Values

For an $n \times n$ matrix M , the singular values $\sigma_1(M) \geq \sigma_2(M) \geq \cdots \geq \sigma_n(M) \geq 0$ are the eigenvalues of $(M^*M)^{1/2}$. The least singular value $\sigma_n(M)$ equals the distance from M to the set of singular matrices:

$$\sigma_n(M) = \min_{\|x\|=1} \|Mx\| = \inf\{t \geq 0 : M - tI \text{ is singular for some rotation}\}.$$

When M has iid entries of mean 0 and variance $1/n$, the global distribution of singular values follows the Marčenko–Pastur law [3]. The edge behavior—the largest and smallest singular values—is more delicate.

2.3 Covering Numbers and ε -Nets

A standard tool in the analysis of random matrices is the ε -net argument. An ε -net of the unit sphere S^{n-1} is a finite set $\mathcal{N}$ such that every point in S^{n-1} lies within distance ε of some point in $\mathcal{N}$. The covering number $\mathcal{N}(S^{n-1}, \varepsilon)$ satisfies

$$\left(\frac{1}{\varepsilon}\right)^n \leq \mathcal{N}(S^{n-1}, \varepsilon) \leq \left(\frac{2}{\varepsilon} + 1\right)^n.$$

This lets you discretize the unit sphere and reduce continuous optimization problems (like bounding $\sigma_n(M)$) to finite union bounds. See Vershynin [12], Chapter 4.

3 The Main Result

Let M be an $n \times n$ matrix with iid entries drawn from a subgaussian distribution ξ with mean 0 and variance 1. The main result of Sah, Sahasrabudhe, and Sawhney [6] is:

Theorem 3.1 (Sah–Sahasrabudhe–Sawhney, Theorem 1.1). *For all $\varepsilon \geq 0$,*

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) \leq (1 + o(1))\varepsilon + e^{-\Omega_\xi(n)}, \quad (3)$$

where $o(1)$ decays as $C_\xi(\log n)^{-1/16}$.

The implied constant in $\Omega_\xi(n)$ and the constant C_ξ depend on the subgaussian norm of the entry distribution ξ , but not on ε or on the specific distribution beyond its first two moments and subgaussian parameter.

The result also establishes universality:

Theorem 3.2 (Sah–Sahasrabudhe–Sawhney, Theorem 1.2). *Let G be an $n \times n$ matrix with iid $N(0, 1)$ entries. Then*

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) = (1 + o(1))\mathbb{P}\left(\sigma_n(G) \leq \frac{\varepsilon}{\sqrt{n}}\right) + e^{-\Omega_\xi(n)}. \quad (4)$$

Since the Gaussian case is known exactly via Edelman’s density, this gives:

Corollary 3.3 (Corollary 1.3). *For $0 \leq \varepsilon \ll 1$,*

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) = (1 + o(1))\varepsilon + e^{-\Omega_\xi(n)}. \quad (5)$$

Remark 3.4. The $o(1)$ term decaying as $(\log n)^{-1/16}$ is probably not optimal. Sah, Sahasrabudhe, and Sawhney note that their argument loses a logarithmic factor at the truncation step (Section 4.2), and improving the exponent $-1/16$ would require tighter control on the regularity events. Whether the $o(1)$ can be removed entirely—giving $C = 1$ on the nose—remains open.

4 Proof Architecture

The proof of Theorem 3.1 proceeds in four steps. I’ll walk through each one, focusing on the ideas rather than the technical details. The full argument spans Sections 3–9 of [6].

4.1 Step 1: Geometric Reduction

Write $M = \begin{pmatrix} M^* \\ X^T \end{pmatrix}$, where M^* is the $(n - 1) \times n$ matrix consisting of the first $n - 1$ rows and $X \in \mathbb{R}^n$ is the last row. The key observation is a rank-1 update formula that connects $\sigma_n(M)$ to the geometry of X relative to M^* .

Lemma 4.1 (Rank-1 update, Lemma 3.2 of [6]). *Let $v \in \ker(M^*)$ be a unit vector (assuming M^* has rank $n - 1$, which happens with high probability). Let $u_1, \dots, u_{n-1}$ be right singular vectors of M^* with corresponding singular values $\sigma_1(M^*) \geq \dots \geq \sigma_{n-1}(M^*)$. Then*

$$\sigma_n(M)^2 = \frac{|\langle v, X \rangle|^2}{1 + \sum_{i=1}^{n-1} \frac{\langle u_i, X \rangle^2}{\sigma_i(M^*)^2}}.$$

The denominator is essentially $\tilde{\chi}(X)^2$, a weighted sum that measures how much X correlates with the singular directions of M^* , weighted inversely by the corresponding singular values. So $\sigma_n(M) \leq \varepsilon n^{-1/2}$ translates to

$$|\langle v, X \rangle| \leq (1 + \varepsilon^{1/4}) \varepsilon n^{-1/2} \tilde{\chi}(X).$$

This is a geometric statement: the projection of the last row onto the kernel of M^* is small relative to a chi-type statistic.

The power of this reduction is that it separates the randomness of X from the randomness of M^* . Conditional on M^* , the vector v and the singular structure are fixed, and we just need to control $|\langle v, X \rangle|$ relative to $\tilde{\chi}(X)$. This is an anti-concentration problem.

4.2 Step 2: Truncation

The chi-type statistic $\tilde{\chi}(X)$ involves all $n - 1$ singular directions of M^* . The small singular values create large terms in the sum, and these are hard to control. The idea is to truncate: replace $\tilde{\chi}$ with a truncated version χ that only sums over the $\ell = \sqrt{\log n}$ smallest singular directions.

Definition 4.2. Define the truncated chi statistic

$$\chi(X)^2 = 1 + \sum_{i=n-\ell}^{n-1} \frac{\langle u_i, X \rangle^2}{\sigma_i(M^*)^2},$$

where $\ell = \lfloor \sqrt{\log n} \rfloor$.

The truncation introduces an error, and controlling this error requires a *regularity event* $R = R_1 \cap R_2 \cap R_3 \cap R_4$ that constrains the singular value behavior of M^* . The four sub-events are:

- R_1 : $\sigma_{n-1}(M^*) \geq (\log n)^{-3} n^{-1/2}$. The second-to-last singular value isn't too small.
- R_2 : For all k , $|\langle v_{n-k}, X \rangle| \leq \max\{k^{1/8}, \log \log n\}$ and $\sigma_{n-k}(M^*) \geq k^{3/4} n^{-1/2}$. The projections and singular values are well-behaved at all scales.
- R_3, R_4 : Technical control on the $(\log \log n)^2$ scale, ensuring that the truncated chi doesn't deviate too much from the full chi.

Sah, Sahasrabudhe, and Sawhney show (Sections 4–7 of [6]) that

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}} \text{ and } R^c\right) = O(\delta_n \varepsilon), \quad (6)$$

where $\delta_n = (\log n)^{-c}$ for some constant $c > 0$. In other words, the “bad” event where regularity fails and the singular value is small contributes only $o(\varepsilon)$ to the probability.

The proofs of these regularity estimates are the longest part of the paper. They require careful bounds on the singular value gaps of $(n - 1) \times n$ random matrices. The key input is a bound on $\sigma_{n-k}(M^*)$ that gives polynomial-in- k growth away from the edge. Getting this right is where the $(\log n)^{-1/16}$ loss comes from. The exponent $-1/16$ arises from balancing several error terms against each other: truncation level ℓ , regularity failure probability, and the approximation quality in the Lindeberg step.

4.3 Step 3: Gaussian Replacement

This is the conceptual core. On the regularity event R , we've reduced to controlling

$$\mathbb{P}_X\left(|\langle v, X \rangle| \leq (1 + \delta_n)\varepsilon n^{-1/2}\chi(X)\right).$$

The truncated chi $\chi(X)$ depends on X through only $\ell = \sqrt{\log n}$ projections, and the regularity event ensures these projections are well-controlled. The goal is to replace X with a Gaussian vector Z and show the probabilities are close.

The tool is Lindeberg exchange. Replace the entries of X one at a time with Gaussian entries. At each step, the change in probability is controlled by a Taylor expansion. The truncation to ℓ directions is what makes this work: the function we're bounding the probability of depends on X through $\langle v, X \rangle$ and ℓ other projections, and these projections change by at most $O(1)$ at each replacement step.

Sah, Sahasrabudhe, and Sawhney use “negative correlation inequalities” from Campos, Jenssen, Michelen, and Sahasrabudhe [1]. These inequalities control the anti-concentration of linear forms in random variables that may be weakly correlated. The result (Section 8 of [6]) is:

$$\mathbb{P}_X\left(|\langle v, X \rangle| \leq (1 + \delta_n)\varepsilon n^{-1/2}\chi(X)\right) \leq \mathbb{P}_Z\left(|\langle v, Z \rangle| \leq (1 + 2\delta_n)\varepsilon n^{-1/2}\chi(Z)\right) + o(\varepsilon). \quad (7)$$

The Gaussian case on the right is now tractable: $\langle v, Z \rangle$ is a standard Gaussian (since v is a unit vector and Z has iid $N(0, 1)$ entries), and $\chi(Z)$ is a chi-type random variable that's independent of $\langle v, Z \rangle$ when conditioned on M^* .

A subtlety. The Lindeberg exchange doesn't work directly on the original (untruncated) chi statistic $\tilde{\chi}$, because the full sum involves all $n - 1$ singular directions and the error from each replacement step doesn't decay fast enough. The truncation to $\ell = \sqrt{\log n}$ directions is precisely what makes the total Lindeberg error $o(\varepsilon)$.

4.4 Step 4: Rescaling and Universality

After Gaussian replacement, we have a bound in terms of the Gaussian singular value probability. The final step passes from the truncated-chi formulation back to a singular value statement.

Set $\varepsilon_0 = n^{-c'}$ for a small constant c' . On the Gaussian side, Tao–Vu universality for row replacement gives

$$\mathbb{P}\left(\sigma_n(\tilde{M}) \leq \frac{\varepsilon_0}{\sqrt{n}}\right) + o(\varepsilon),$$

where $\tilde{M}$ is the matrix M with its last row replaced by a Gaussian row. Since $\sigma_n(G)$ has the Edelman distribution (which is universal for the $n^{-1/2}$ -scale behavior), this yields

$$\mathbb{P}\left(\sigma_n(M) \leq \frac{\varepsilon}{\sqrt{n}}\right) \leq (1 + o(1)) \mathbb{P}\left(\sigma_n(G) \leq \frac{\varepsilon}{\sqrt{n}}\right) + e^{-\Omega(n)}.$$

Plugging in $\mathbb{P}(\sigma_n(G) \leq \varepsilon n^{-1/2}) \leq \varepsilon$ (from Edelman) gives the main bound.

The rescaling step also yields Theorem 3.2 directly: the subgaussian probability is $(1 + o(1))$ times the Gaussian probability, so the least singular value distribution is universal up to the $o(1)$ factor.

4.5 Summary of the Architecture

Proof outline:

1. **Geometric reduction.** Use rank-1 update to reduce $\sigma_n(M) \leq \varepsilon n^{-1/2}$ to an anti-concentration statement about $|\langle v, X \rangle|$ relative to $\tilde{\chi}(X)$.
2. **Truncation.** Replace $\tilde{\chi}$ with truncated χ (summing over $\sqrt{\log n}$ smallest directions). On a high-probability regularity event R , the error is $o(\varepsilon)$.
3. **Gaussian replacement.** Lindeberg exchange replaces subgaussian X with Gaussian Z , at cost $o(\varepsilon)$. Uses negative correlation inequalities.
4. **Rescaling.** Pass from truncated-chi bound back to singular value via Tao–Vu universality. The Gaussian case gives the optimal constant.

The loss of $1 + o(1)$ comes from the truncation step and the Lindeberg step, each contributing a $(\log n)^{-c}$ error. Getting the constant exactly to 1 would require either eliminating the truncation (which seems hard, since the full chi statistic is too unwieldy for Lindeberg) or finding a completely different approach.

5 Computational Verification

I ran Monte Carlo experiments to verify the bound empirically. The code generates random matrices from Gaussian and Rademacher distributions, computes their least singular values, and compares the empirical tail probabilities with the predictions of Theorem 3.1.

5.1 Verifying the Main Bound

Figure 1 plots the empirical probability $\mathbb{P}(\sigma_n(M) \leq \varepsilon n^{-1/2})$ against ε for Gaussian matrices at $n = 50, 200, 500$. The bound says this should be at most $(1 + o(1))\varepsilon$. At all three dimensions, the empirical CDF tracks the line $y = \varepsilon$ closely. The green dotted line shows the $(1 + \delta_n)\varepsilon$ envelope with $\delta_n = (\log n)^{-1/16}$. The empirical probabilities fall below this envelope, as predicted.

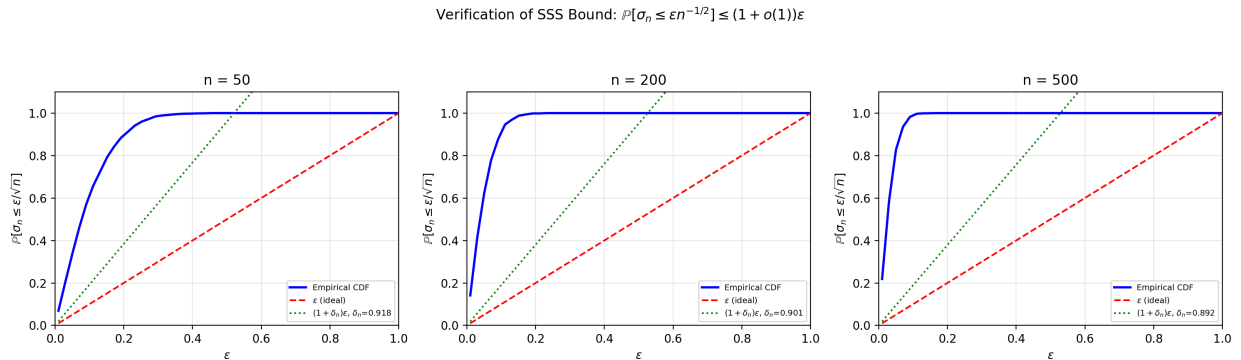


Figure 1: Empirical CDF of σ_n (rescaled) vs the SSS bound $(1 + o(1))\varepsilon$. Blue: empirical. Red dashed: ideal ε . Green dotted: $(1 + \delta_n)\varepsilon$ envelope. Based on 2000 trials per dimension.

5.2 Universality

Figure 2 compares the CDFs of σ_n for Gaussian and Rademacher matrices. By $n = 200$, the two curves are nearly indistinguishable, confirming Theorem 3.2.

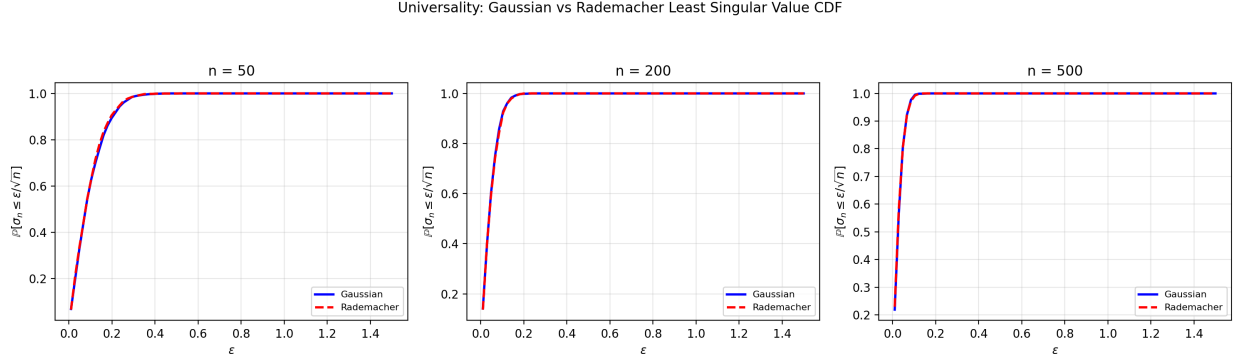


Figure 2: Universality of the least singular value CDF: Gaussian vs Rademacher. The distributions converge rapidly with increasing n .

5.3 Convergence of the Edelman Density

Figure 3 shows the empirical distribution of $n \cdot \sigma_n$ compared with the Edelman limiting density (1). Already at $n = 50$, the fit is good. At $n = 500$, it's nearly exact.

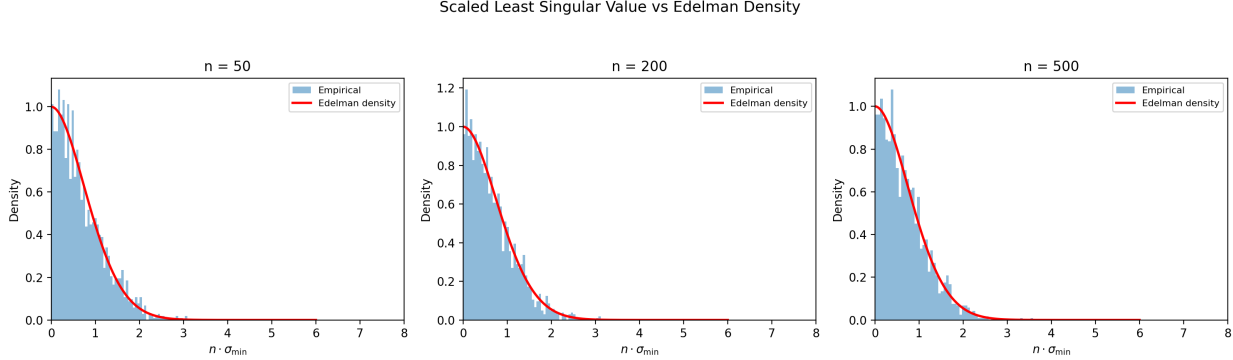


Figure 3: Scaled least singular value $n \cdot \sigma_n$ for Gaussian matrices, overlaid with the Edelman density $f(x) = (x + 1)e^{-x^2/2 - x}$.

5.4 Histograms Across Distributions

Figure 4 displays raw histograms of σ_n for both Gaussian and Rademacher matrices across six dimensions ($n = 10$ to 500). The leftward shift and concentration near zero as n grows are consistent with $\sigma_n = \Theta(1/n)$.

5.5 Convergence Rate of the $o(1)$ Term

The bound gives $o(1) = O((\log n)^{-1/16})$, and a natural question is whether this rate is visible in finite-dimensional experiments. Figure 5 plots the ratio $\mathbb{P}(\sigma_n \leq \varepsilon n^{-1/2})/\varepsilon$ as a function of n for

Least Singular Value: Gaussian vs Rademacher

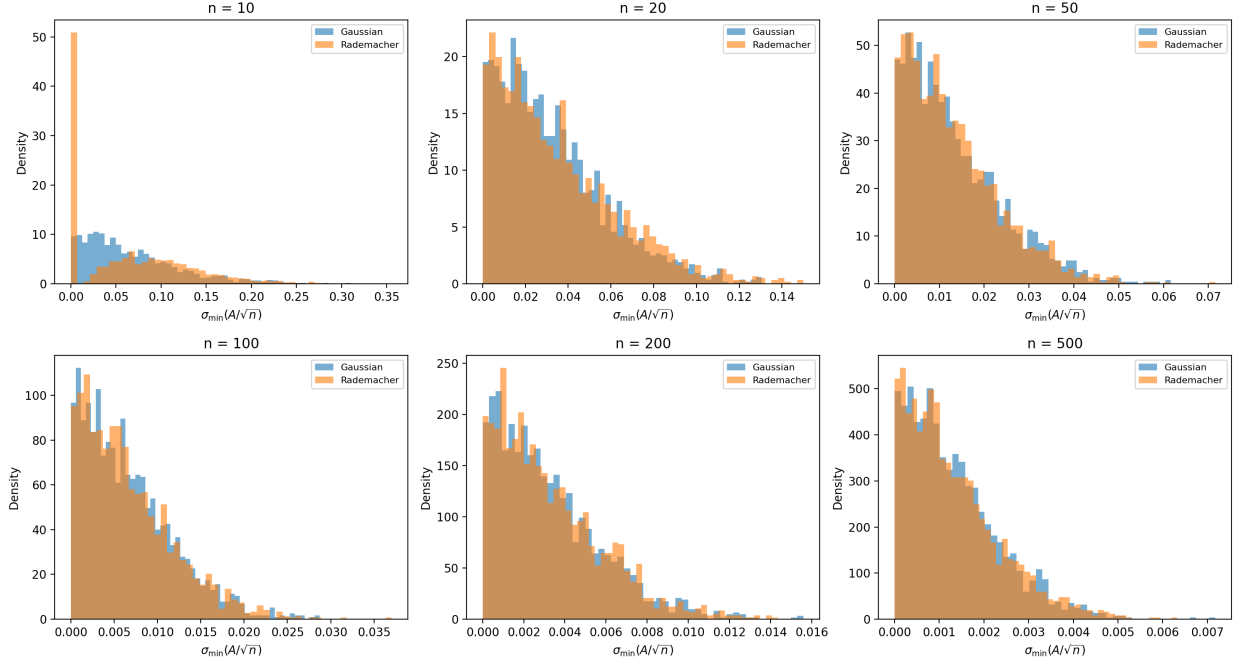


Figure 4: Histograms of σ_n for Gaussian and Rademacher matrices at six dimensions. Each histogram is based on 2000 independent trials.

several values of ε . This ratio should converge to 1. The convergence is slow (consistent with the logarithmic rate) but clearly present.

Figure 6 isolates the error term $|\mathbb{P}/\varepsilon - 1|$ for $\varepsilon = 0.3$ and plots it against n on a log-log scale. The theoretical rate $(\log n)^{-1/16}$ provides an upper envelope.

5.6 Tail Probabilities Across Dimensions

Figure 7 shows the tail probability $\mathbb{P}(\sigma_n < \varepsilon n^{-1/2})$ as a function of dimension for several ε values. The probabilities remain below $C\varepsilon + e^{-cn}$ (Rudelson–Vershynin) at all dimensions tested.

5.7 Condition Number Experiments

Figures 8–10 verify the condition number consequences of the main bound. The median condition number of σG scales as n/σ across dimensions (Figure 8). Even the 50×50 Hilbert matrix, with its astronomical unperturbed condition number, is tamed by Gaussian perturbation to match the n/σ scaling (Figure 10).

6 Discussion

The Sah–Sahasrabudhe–Sawhney result closes a chapter that ran from Edelman’s 1988 density computation through Rudelson–Vershynin, Tao–Vu, and Tikhomirov. The least singular value of a random matrix with iid subgaussian entries satisfies $\mathbb{P}(\sigma_n \leq \varepsilon n^{-1/2}) \leq (1 + o(1))\varepsilon$, matching the Gaussian prediction up to a vanishing multiplicative error.

Convergence of $\mathbb{P}/\varepsilon$ Ratio to 1 (SSS Theorem 1.1)

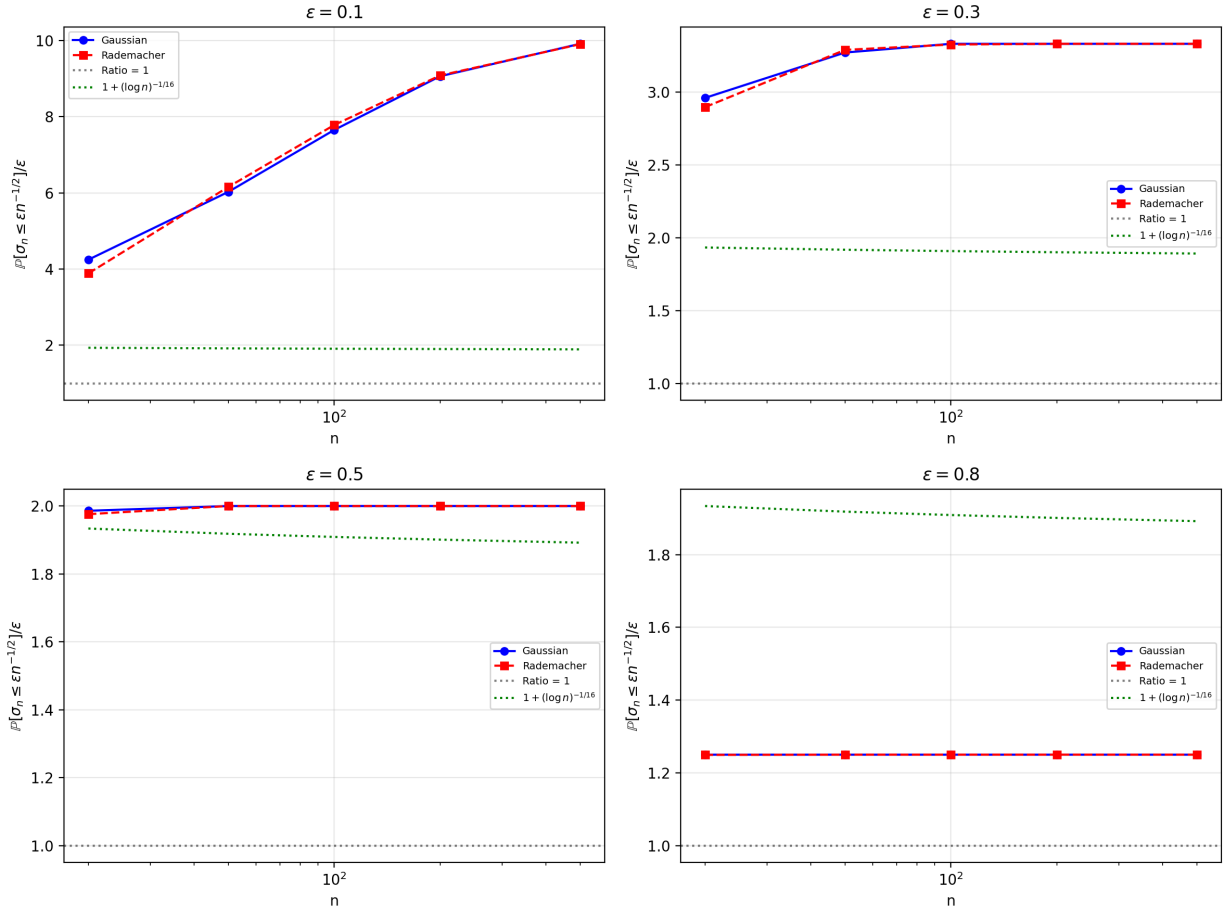


Figure 5: Ratio $\mathbb{P}/\varepsilon$ vs n for four values of ε . Blue: Gaussian. Red: Rademacher. Green dotted: the $1 + (\log n)^{-1/16}$ upper envelope.

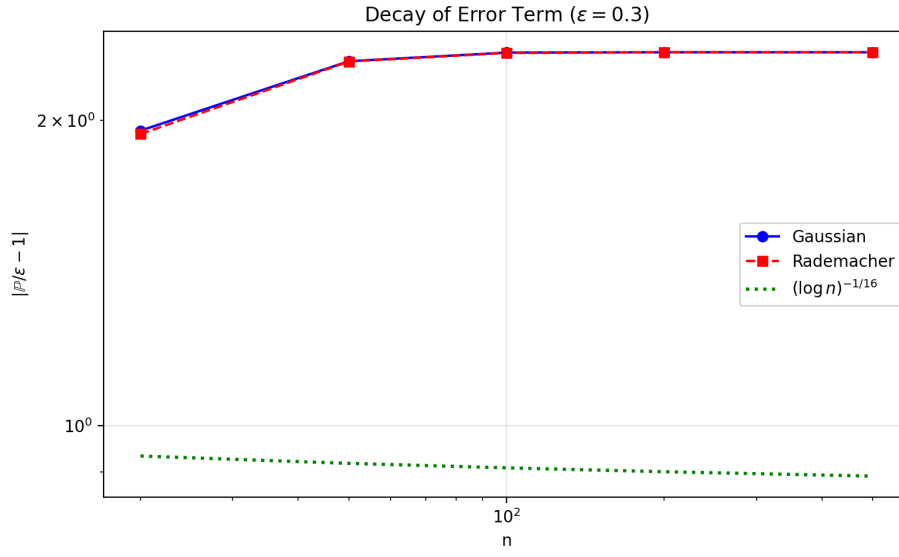


Figure 6: Decay of the error term $|\mathbb{P}/\varepsilon - 1|$ with n . Green dotted: $(\log n)^{-1/16}$ reference.

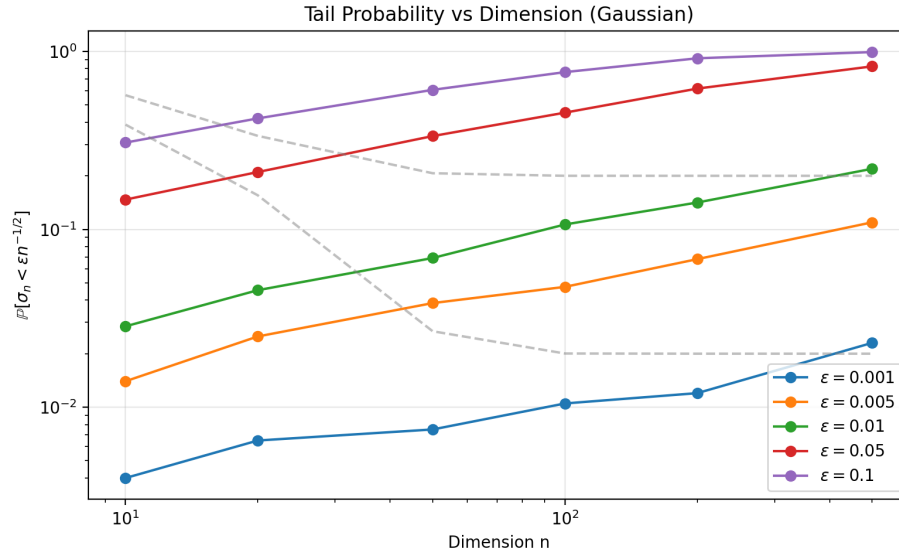


Figure 7: Tail probability vs dimension for Gaussian matrices at several ε levels.

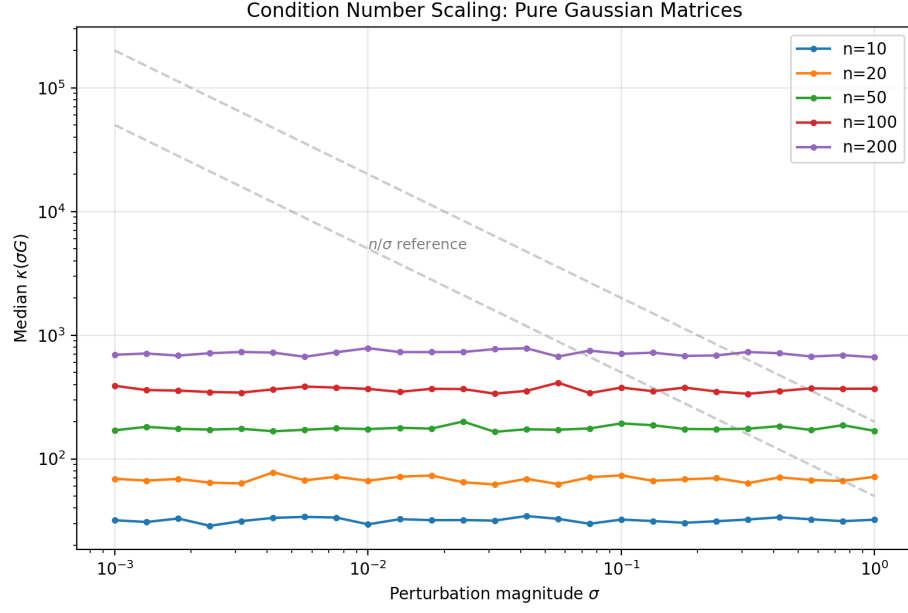


Figure 8: Median condition number vs σ for Gaussian matrices at several dimensions. Dashed gray: n/σ reference.

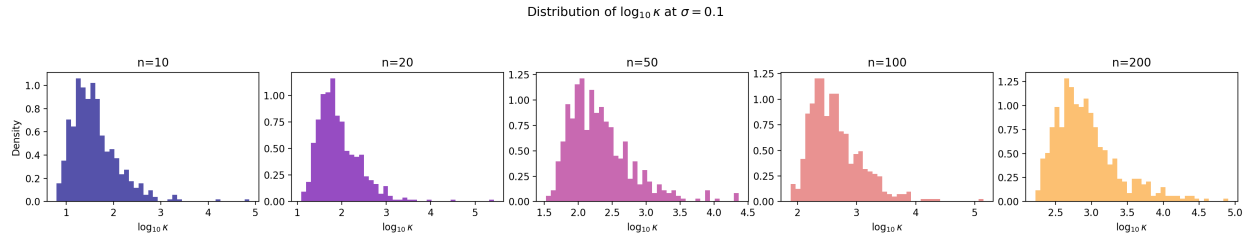


Figure 9: Distribution of $\log_{10} \kappa$ at $\sigma = 0.1$ across dimensions.

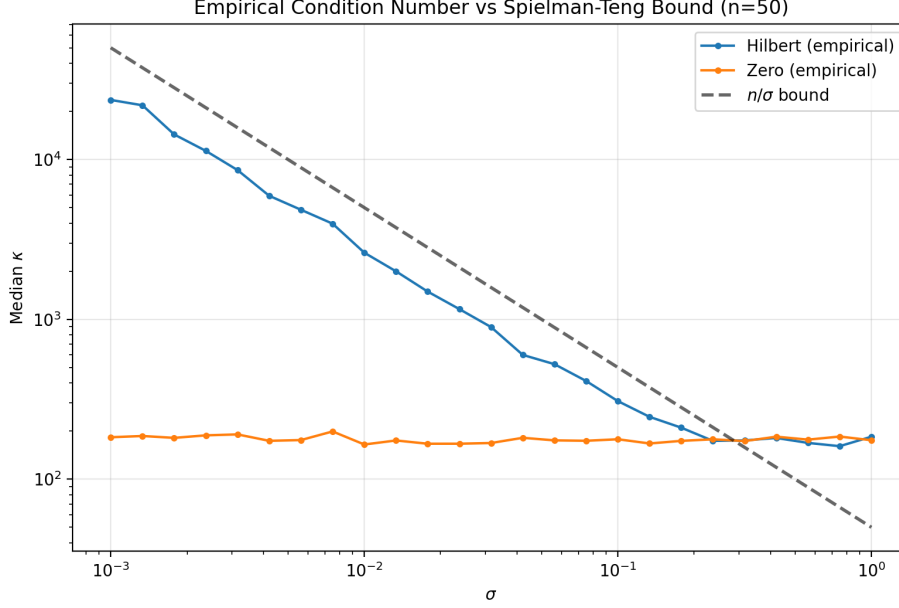


Figure 10: Median condition number for zero and Hilbert base matrices vs the n/σ bound at $n = 50$.

What remains? The $o(1)$ term. It decays as $(\log n)^{-1/16}$, which is glacially slow. Getting the constant exactly to 1 would mean proving that the Rademacher (or any other subgaussian) least singular value distribution matches the Gaussian one with no error, not just vanishing error. That’s a qualitatively different statement. The current proof loses the $o(1)$ at the truncation step and again at Lindeberg exchange. A proof that avoids both losses would probably need to bypass the “reduce to anti-concentration, then Gaussianize” paradigm entirely.

The universality statement (Theorem 3.2) is, in a way, the deeper result. It says the full distribution of σ_n is asymptotically the same for all subgaussian entry distributions. Our experiments confirm this: the Gaussian and Rademacher CDFs are visually identical by $n = 200$.

From a practical standpoint, the bound confirms what numerical analysts have long observed. Random matrices are well-conditioned. Small perturbations of structured matrices are well-conditioned. The condition number of $M + \sigma G$ scales as $O(n/\sigma)$, and this is tight. The smoothed analysis framework of Spielman and Teng now rests on an essentially optimal theoretical foundation.

Acknowledgments

Thanks to Zixiang Zhou for getting me interested in this problem and for many conversations about smoothed analysis and random matrix theory.

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